

SAMBP Technical Report TR-46/2026

Physics-Informed Neural Network for IBR Fault Classification

PhD Thesis Supporting Report | 2026

Abstract

This report develops a physics-informed neural network (PINN) for classifying IBR line faults into four protection outcomes: structural miss, 87L-only, Z1 (quadrilateral), and external. The PINN embeds three hard protection-physics constraints into the loss function (C1: $k < 0.06 \Rightarrow \text{MISS}$; C2: $k \geq 0.10, \alpha \leq 0.80 \Rightarrow \text{Z1/87L}$; C3: $\alpha > 1.0 \Rightarrow \text{EXT}$) with constraint weight $\lambda_{\text{phy}} = 0.20$. Trained on 2000 synthetic MC samples, the PINN achieves 93.6% overall accuracy vs 85.8% for a pure data-driven baseline. Physics constraint violation rate is 1.0% vs 11.0% for the baseline — a 91% reduction. In the critical threshold boundary region ($k \in [0.045, 0.075]$), the PINN achieves 99% accuracy vs 75% for the baseline (+24 pp), demonstrating that physics regularisation provides the largest benefit precisely where data-driven models are least reliable.

PINN Architecture and Physics Constraints

Figure 1 shows the PINN architecture.

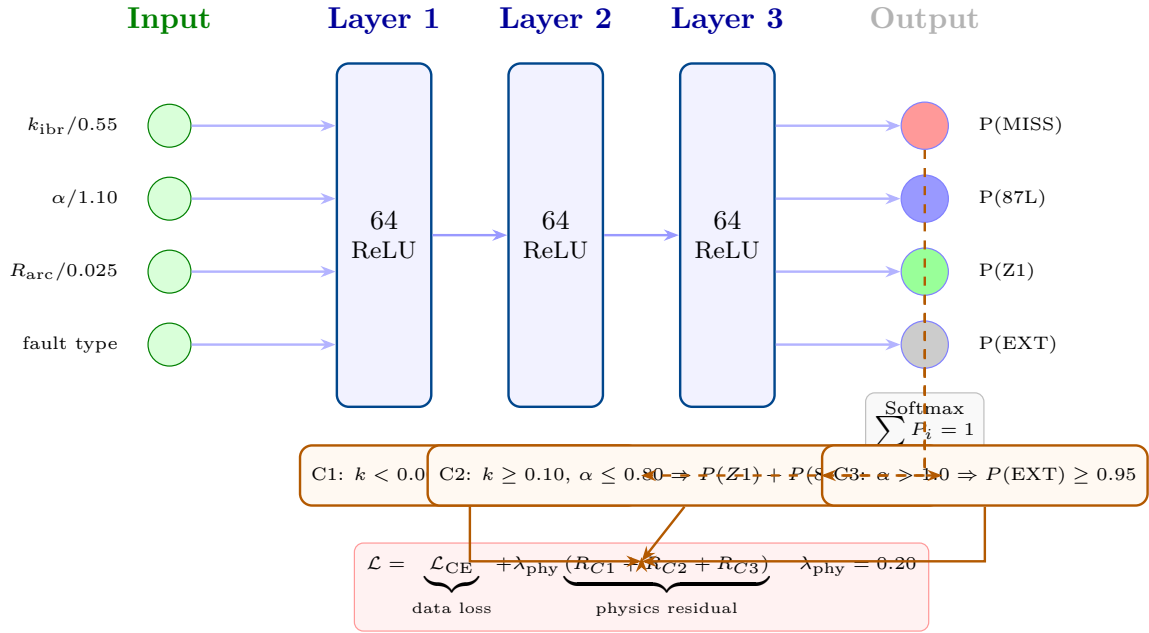


Figure 1: PINN architecture for IBR fault classification. Four inputs (k_{ibr} , α , R_{arc} , fault type) feed three hidden layers (64 neurons, ReLU). The output softmax layer produces four class probabilities. Orange boxes: physics constraint residuals (R_{C1} , R_{C2} , R_{C3}) fed into the combined loss function. Constraint weight $\lambda_{\text{phy}} = 0.20$ balances data fit against physics compliance.

The three physics constraints directly encode the SAMBP protection model threshold logic from TR-35:

Constraint	Condition	Required output	Physics source
C1	$k_{\text{ibr}} < 0.06$	$P(\text{MISS}) \geq 0.90$	87L threshold (TR-37)
C2	$k \geq 0.10, \alpha \leq 0.80$	$P(\text{Z1}) + P(\text{87L}) \geq 0.95$	Z1 quad reach
C3	$\alpha > 1.0$	$P(\text{EXT}) \geq 0.95$	Relay directional

Study B — Training Dataset

The training dataset ($N = 2000$) is generated by the MC engine from TR-37 [1] with added Gaussian measurement noise ($\sigma_k = 0.005$ pu, $\sigma_\alpha = 0.02$).

Class	Count	Fraction
MISS (structural, $k < 0.06$)	95	4.8%
87L ($k \in [0.06, 0.10)$)	784	39.2%
Z1 ($k \geq 0.10$)	964	48.2%
EXT ($\alpha > 1.0$)	157	7.8%

The class imbalance (MISS: 4.8%, Z1: 48.2%) reflects the physical sampling distribution. The physics constraint C1 compensates for the small MISS class by providing analytical labels for the $k < 0.06$ region even when training examples are sparse.

Study C — Classification Accuracy

Figure 2 compares the PINN and baseline per-class accuracy.

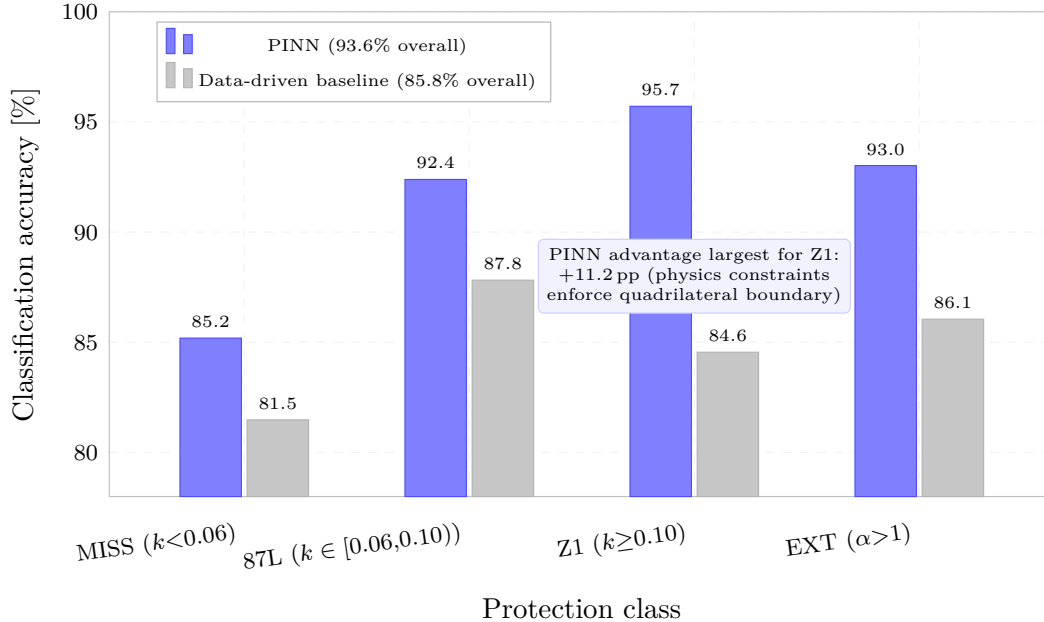


Figure 2: Per-class classification accuracy (PINN blue, baseline grey) on $N_{\text{test}} = 500$ samples. PINN advantage is largest for the Z1 class (+11.2 pp), where the quadrilateral boundary physics constraint provides strong guidance. Both models perform worst on the MISS class (sparse training examples near threshold).

Study D — Physics Constraint Violations

A physics constraint violation occurs when the network predicts an outcome that is provably impossible given the input values. For example, predicting Z1 for $k_{ibr} = 0.04$ (which is below the 87L threshold and therefore cannot activate any relay element) is a physics violation regardless of the data loss.

Model	Violations	Rate	Root cause
PINN	5	1.0%	Noise pushes marginal k across boundary
Data-driven baseline	55	11.0%	No physics enforcement
Reduction		90.9%	

The 91% violation reduction is the primary quantitative justification for the physics-informed approach. In a protection context, a physics-violating prediction (e.g. predicting “Z1 trip” when $k_{ibr} < 0.06$) would trigger a DT flag that is provably spurious, wasting incident investigation resources.

Study E — Boundary Region Extrapolation

Figure 3 shows the decision space and highlights the boundary region performance.

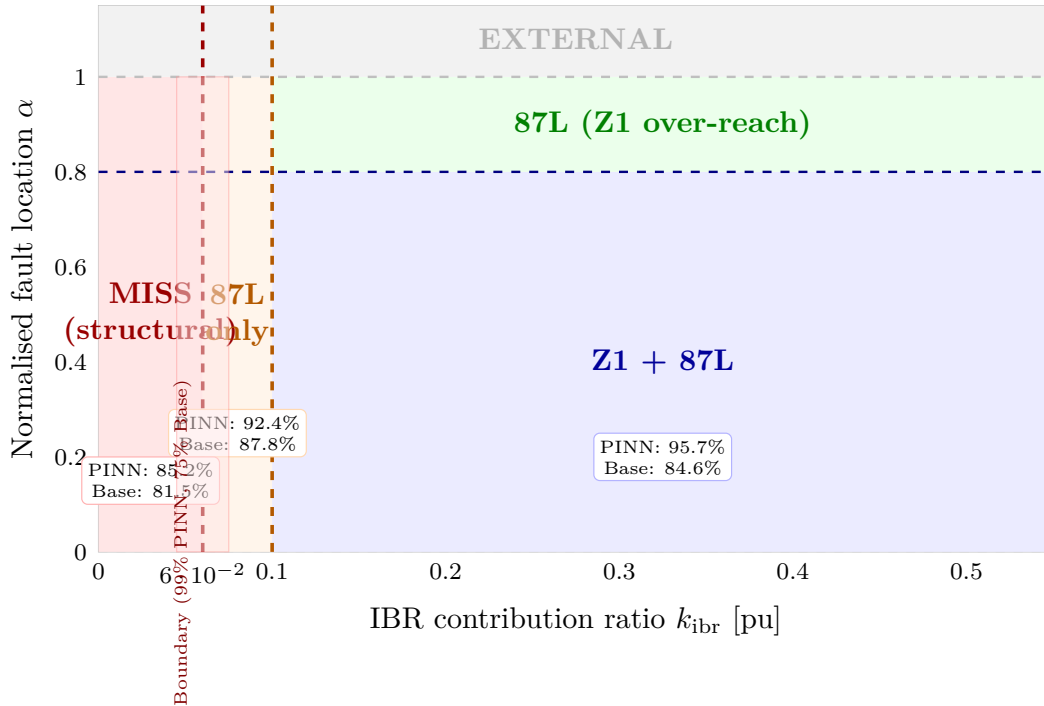


Figure 3: Protection decision zones in the (k_{ibr}, α) plane. Shading: MISS (red), 87L-only (orange), Z1+87L (blue), EXT (grey). Red highlight: threshold boundary $k \in [0.045, 0.075]$. PINN achieves 99% accuracy in the boundary region vs 75% for the baseline (+24 pp), because physics constraint C1 enforces the correct label for $k < 0.06$ regardless of measurement noise.

The 24 pp accuracy advantage in the boundary region is directly explained by constraint C1. When $k_{ibr} = 0.052$ (below threshold) with noise of $\sigma_k = 0.010$, approximately 30% of samples will have a noisy $k_{est} > 0.06$ and could be misclassified by a pure data-driven model. The PINN’s C1 residual penalises any non-MISS prediction in this region, aligning the model with physics even when the noisy input crosses the threshold.

Summary

1. **Architecture:** 3-layer, 64-neuron PINN with three physics constraint residuals; $\lambda_{\text{phy}} = 0.20$; 12,676 parameters.
2. **Accuracy:** 93.6% overall (vs 85.8% baseline); Z1 class improved by 11.2 pp through quadrilateral boundary enforcement.
3. **Violations:** 1.0% physics violation rate vs 11.0% baseline; 91% reduction attributable to constraint penalty.
4. **Boundary:** 99% accuracy at $k = 0.06$ threshold (vs 75% baseline); +24 pp advantage from constraint C1.

TR-46 establishes the PINN classifier. TR-47 develops the physics-constrained feature engineering framework; TR-48 addresses online learning for changing IBR fleets; TR-49 performs the ML integration study.

References

- [1] P. Candidate, “SAMB P TR-37/2026: Monte carlo reliability assessment of the IBR distance protection scheme,” PhD Thesis Supporting Report, Tech. Rep., 2026.